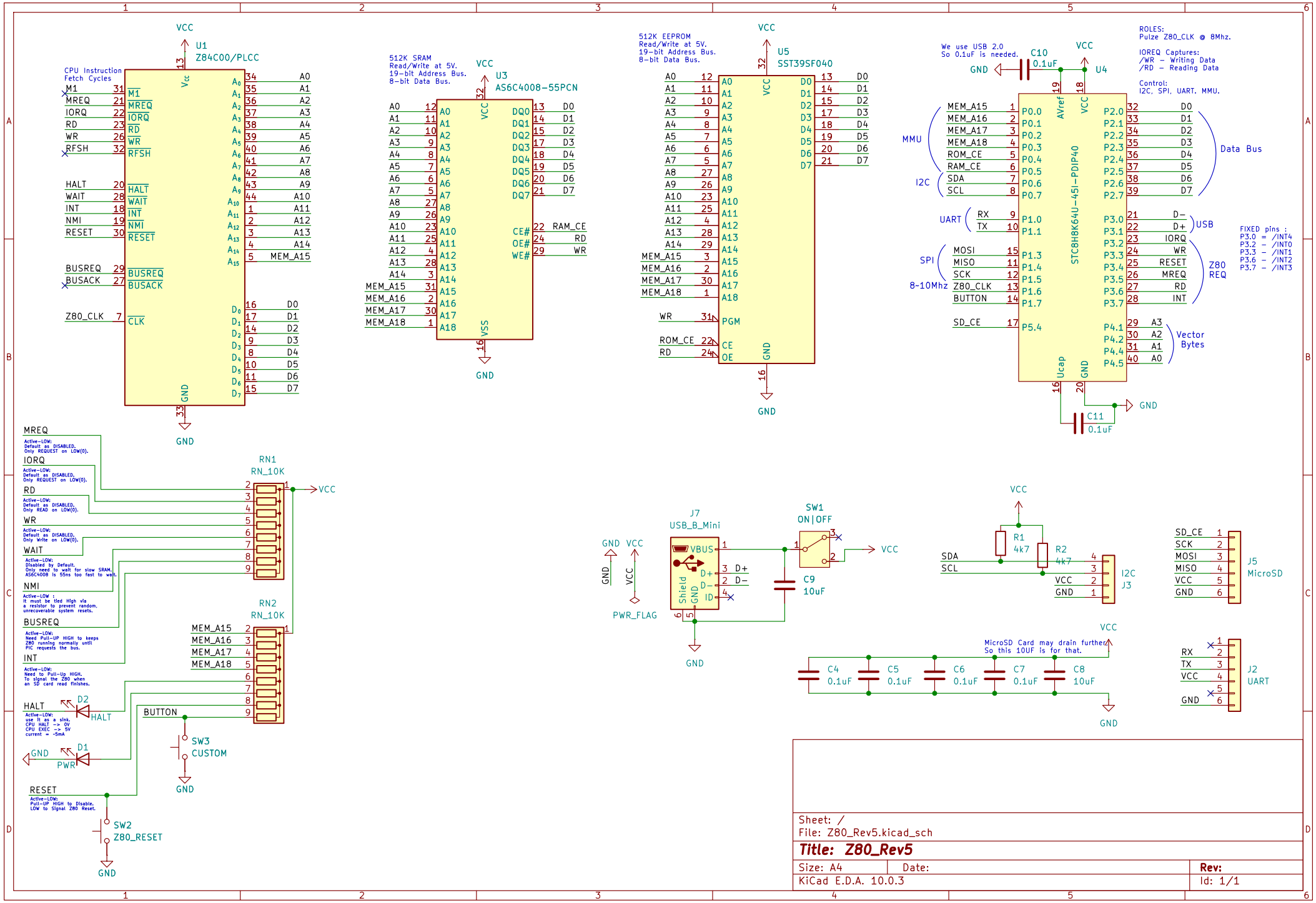
U1
Z84C00/PLCC
CPU Instruction Fetch Cycles
M1
MREQ
IORQ
RD
WR
RFSH
HALT
WAIT
INT
NMI
RESET
BUSREQ
BUSACK
Z80_CLK
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
MEM_A15
D0
D1
D2
D3
D4
D5
D6
D7
VCC
GND
U3
AS6C4008-55PCN
512K SRAM
Read/Write at 5V.
19-bit Address Bus.
8-bit Data Bus.
CE#
OE#
WE#
RAM_CE
RD
WR
U5
SST39SF040
512K EEPROM
Read/Write at 5V.
19-bit Address Bus.
8-bit Data Bus.
PGM
ROM_CE
U4
STC8H8K64U-45I-PDIP40
We use USB 2.0
So 0.1uF is needed.
C10
0.1uF
C11
0.1uF
ROLES:
Pulze Z80_CLK @ 8Mhz.
IOREQ Captures:
/WR - Writing Data
/RD - Reading Data
Control:
I2C, SPI, UART. MMU.
MMU
I2C
UART
SPI
8-10Mhz
Data Bus
USB
Z80 REQ
Vector Bytes
FIXED pins :
P3.0 = /INT4
P3.2 = /INT0
P3.3 = /INT1
P3.6 = /INT2
P3.7 = /INT3
RN1
RN_10K
RN2
RN_10K
Active-LOW:
Default as DISABLED.
Only REQUEST on LOW(0).
Active-LOW:
Default as DISABLED.
Only READ on LOW(0).
Active-LOW:
Default as DISABLED.
Only Write on LOW(0).
Active-LOW:
Disabled by Default.
Only need to wait for slow SRAM.
AS6C4008 is 55ns too fast to wait.
Active-LOW :
It must be tied High via
a resistor to prevent random,
unrecoverable system resets.
Active-LOW:
Need Pull-UP HIGH to keeps
Z80 running normally until
PIC requests the bus.
Active-LOW:
Need to Pull-Up HIGH.
To signal the Z80 when
an SD card read finishes.
Active-LOW:
use it as a sink.
CPU HALT -> 0V
CPU EXEC -> 5V
current = ~5mA
Active-LOW:
Pull-UP HIGH to Disable.
LOW to Signal Z80 Reset.
D1
PWR
D2
HALT
SW2
Z80_RESET
SW3
CUSTOM
J7
USB_B_Mini
VBUS
D+
D-
ID
Shield
PWR_FLAG
SW1
ON|OFF
C9
10uF
R1
4k7
R2
4k7
I2C
J3
SD_CE
SCK
MOSI
MISO
J5
MicroSD
MicroSD Card may drain further.
So this 10UF is for that.
C4
0.1uF
C5
0.1uF
C6
0.1uF
C7
0.1uF
C8
10uF
RX
TX
J2
UART
Sheet: /
File: Z80_Rev5.kicad_sch
Title: Z80_Rev5
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1